Fundamentals of Artificial Intelligence (IN2406)

Programming Exercise I: Search-based Motion Planning with Motion Primitives

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1 Introduction

Finding safe motions in complex traffic situations is one of the major challenges in autonomous driving. Typically, one has to develop a large software framework before one can start developing new motion planning techniques. The CommonRoad benchmark suite provides a large scenarios database and a variety of software tools for developing motion planning algorithms. An exemplary CommonRoad scenario taken from the city center of Munich (Stachus) is shown in Fig.1. CommonRoad also provides a ranking system to which one

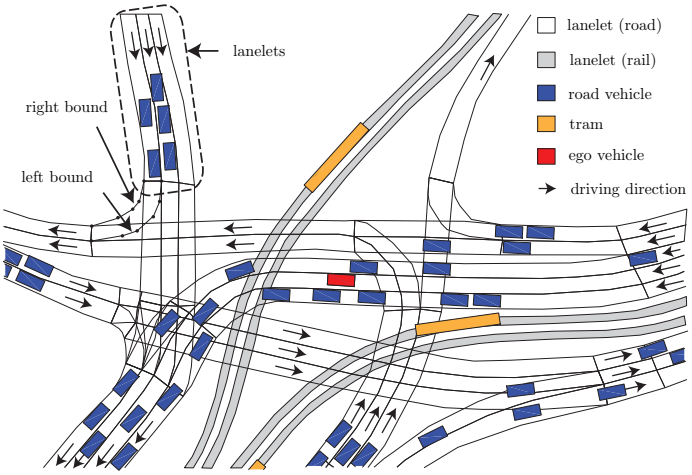


Figure 1: Visualization of Stachus in CommonRoad

can upload his/her solutions to the motion planning problems, and see how well their planner performs in comparison to others.

Your task in this exercise is to implement a heuristic function and/or a search algorithm for a motion planner with motion primitives to solve real traffic scenarios in CommonRoad.

2 Installation

The software for the exercise is located in this repository. There are 3 ways to run the software framework: **Local installation**, **Virtual Machine** or **Docker**.

2.1 Local Installation

To install the software locally, you can follow the instructions given in the `README.md`. The local installation is recommended for platforms **Ubuntu 18.04/20.04** and **MacOS 10.13** or higher. For Windows, we recommend using the other provided options.

2.2 Virtual Machine

A VirtualBox image of Lubuntu 18.04 is provided, in which all necessary packages are installed (requires ca. 18 GB space after unzipping). It can be downloaded via this link. The downloading and the default login password are both **commonroad**. In the terminal, use the command `jupyter notebook` to launch

the Jupyter notebook, and navigate to `/Desktop/commonroad-search/`. If necessary, you can adjust the resolution in **Preferences >Monitor settings >Resolution**.

2.3 Docker

A Docker file and a prebuilt Docker image for this exercise are available (see `docker/README.md` for details).

3 CommonRoad Scenarios

3.1 Scenario

Before implementing the search algorithms, it is essential to correctly understand the structure of CommonRoad scenario files. The most important elements in a Scenario are:

- **Lanelet network:** The road network consists of a series of road segments, which are called lanelets. A lanelet is defined by its left and right bounds, where each bound is represented by an array of points (a polyline), see Fig. 2.
- **Obstacles:** A scenario typically contains both static and dynamic obstacles. Static obstacles do not move over time (e.g., parked vehicles); dynamic obstacles are other traffic participants (e.g., cars, pedestrians, etc.). All obstacles are modeled as geometric shapes. In addition, dynamic obstacles have a trajectory, which models their movement over time. In this exercise, the future trajectories of the dynamic obstacles are given.

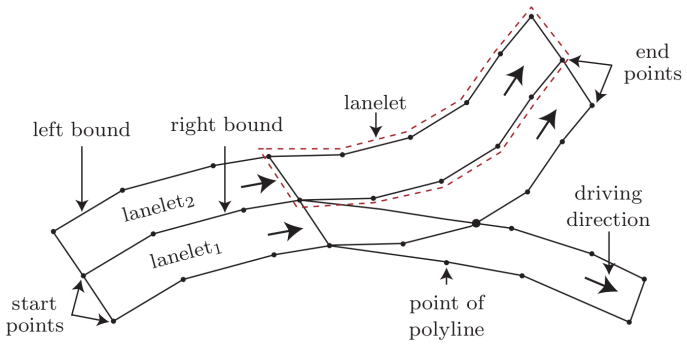


Figure 2: Example of a lanelet network.

3.2 Planning Problem

A motion planning problem is defined by two main attributes, an initial state and a goal region:

- **Initial state:** The initial state of the ego vehicle. Some of its attributes are:
  - **Position:** A two-dimensional vector $[x, y]^{\mathrm{T}}$ in [m].
  - **Steering Angle:** A scalar δ in [rad].
  - **Velocity:** A scalar v in [m/s].
  - **Orientation:** A scalar $\theta \in [-\pi, \pi]$ in [rad].
- **Goal region:** The goal states which should be reached by the ego vehicle. Some of its attributes are:
  - **Position Region:** A two-dimensional polygon within $\mathbb{R}^2$; each axis has the unit [m].
  - **Orientation Interval:** An interval of allowed orientation in [rad].

  - **Velocity Interval:** An interval of allowed velocities in [m/s].
  - **Time Interval:** An interval of allowed time of arrival at the goal region in [time step].

Note that all time stamps in CommonRoad are represented by time steps $k \in \mathbb{N}^+$. The time in seconds is implicitly given by the fixed, global time step size r (usually set to 0.1 second) of the scenario, so that $t = kr$. For example, time step $k = 61$ corresponds to time $t = 6.1$s, given that r is set to 0.1s.

Based on the types of the goal states, the planning problems (and the scenarios) can be categorized into:

- **Regular**: goal states contain explicit requirements on positions, velocities, orientations, and time steps
- **Survival**: goal states only contain requirements on time steps

In survival scenarios, you only need to find a solution trajectory that does not collide with other traffic participants up to the indicated time intervals.

3.3 Solution

A valid solution trajectory to a planning problem satisfies the following conditions:

- It starts at the initial state of the planning problem.
- It reaches one of the goal states of the planning problem.
- The trajectory is drivable by a given vehicle model and vehicle type. In this exercise, we assume that the **Kinematic Single-Track model** is used together with **vehicle type BMW320i**. See Vehicle Models for more details.
- The ego vehicle does not collide with any obstacles and stays on the road while following the trajectory.

4 Software Description

4.1 Code Structure

The source code is structured as follows:

- `SMP/`:
  - `maneuver_automaton/`: code related to motion primitives and maneuver automata.
  - `motion_planner/`: code related to motion planners.
  - `batch_processing/`: code related to batch processing for parallel execution.
- `tutorials/`: essential tutorials for this programming exercise.
- `scenarios/`: scenarios to be solved by the motion planners.
- `outputs/`: folder to hold the outputs (solution files, log files, GIFs, etc.).

4.2 Motion Primitives

Motion primitives are short trajectory segments that can be driven by a given vehicle model. By concatenating the primitives, a drivable trajectory can be constructed that leads the vehicle from the initial state to the goal state. Fig. 3 shows an exemplary search tree that is spanned via motion primitives, where the initial and final states of primitives are indicated by dots. The path between two successive dots are the intermediate states of a primitive.

In this exercise, all of the generated primitives start at position $[x = 0, y = 0]^{\mathrm{T}}$, have a duration of 0.5 seconds, and differ in velocity, steering angle, orientation, acceleration, traveled distance, etc. If the velocity and the steering angle of the *initial* state of a *successor* primitive are equal to those of the *final* state of a *predecessor* primitive, we say that these two primitives are *connectable*.

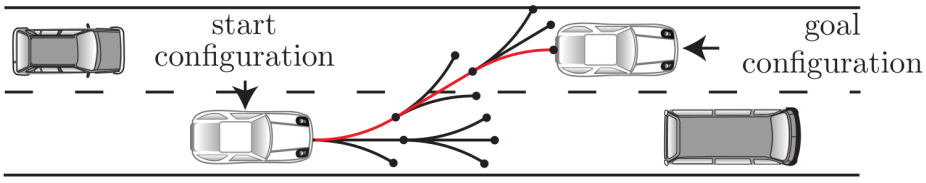


Figure 3: A search tree with motion primitives.

4.3 Motion Planners

We already provide the implementation of the six basic (un)informed search algorithms from the lecture for the motion planning problem:

1. **Breadth First Search**
2. **Depth First Search**
3. **Depth-limited Search**
4. **Uniform Cost Search (aka Dijkstra's algorithm)**
5. **Greedy Best First Search**
6. **A* Search**

Their implementations can be found in `SMP/motion_planner/search_algorithms/`. The individual planner classes are derived from a base class `SearchBaseClass`. Similarly, new search algorithms can be derived from this base class.

4.4 Helpful Tools

We provide some helpful tools that may assist you with designing your motion planner.

4.4.1 Helper Functions

We provide a set of helper functions for heuristic cost computation within the base class `SearchBaseClass`. You do not have to use all of the provided functions, and of course you are free to come up with your own ideas.

4.4.2 DrPlanner

You can also check out our tool DrPlanner, which is the first framework to autonomously diagnose and repair motion planners using the power of large language models.

4.5 Tutorials

We provide the following tutorials to get started with this exercise:

- `0_commonroad_input-output`: this is the first tutorial on CommonRoad Input-Output, and shows how CommonRoad XML-files can be read, modified, visualized, and saved.
- `1_search_algorithms`: this exemplifies how one can combine motion primitives with (un)informed search algorithms to find solutions to simple motion planning problems.
- `2_commonroad_search`: this demonstrates how one can use the (un)informed search algorithms, combined with motion primitives, to solve real motion planning problems introduced in CommonRoad.
- `3_motion_primitive_generator`: this helps you understand how the motion primitives are generated, in case you want to generate your own set of motion primitives.

- `4_batch_processing`: this shows how the search for solutions can be boosted via parallel computation with multi-threading using the given batch processing script. **You can run this script and call your planner to generate solution files for submission.**
- `5_route_planner`: this explains how the route planner package can be used to plan high-level routes for CommonRoad scenarios to better guide the search tree.

4.6 Getting Started

To get started, we suggest going through all of the tutorials mentioned above. After that, you may look into the source code for implementation details in the following order:

1. `SMP/maneuver_automaton/motion_primitive.py`
2. `SMP/maneuver_automaton/maneuver_automaton.py`
3. `SMP/motion_planner/motion_planner.py`
4. `SMP/motion_planner/search_algorithms/`

You should implement your own code in the file `SMP/motion_planner/search_algorithms/student.py`, where we have provided a raw structure which you can fill out or extend. Additionally, we provide an example for an implementation in `SMP/motion_planner/search_algorithms/student_example.py`.

5 Task & Evaluation

Your task is to implement a heuristic function and/or a search algorithm for a motion planner with motion primitives to solve CommonRoad scenarios. You can either:

- improve the heuristic function of the already provided search algorithms (see Sec. 4.3)
- or implement your own search algorithm.

To pass this exercise, you should solve at least **350 out of the 500** given scenarios. Note that even if you produce more solutions to one scenario, it will still be considered as solving one scenario only. Don't be scared, roughly 330 scenarios are already solvable with the provided out-of-the-box planners :)

6 Tips on Heuristic Function

Searching blindly in a high-dimensional state space can be very resource-consuming (time, memory, computing power, etc.); on the other hand, having a well-thought-out heuristic function can dramatically reduce the need for such resources. In general, we want to reward states that lead us closer to the goal state while obeying some constraints. The goal region of the planning problem mostly consists of state components, including position, velocity, orientation, etc., we can therefore try to formulate a heuristic function that penalizes large deviation from the goal state by considering:

- **positional distance:** the positional distance (in Euclidean, Manhattan, or other distances) between the (x, y) position of a given state and the goal region (e.g., the center of the goal region).
- **velocity difference:** the velocity difference between the velocity of a given state and the goal state (e.g., center of the desired velocity interval).
- **orientation difference:** the orientation difference between the orientation of a given state and the goal state (e.g., center of the desired orientation interval).
- **time difference:** the time difference between the time step of a given state and the goal state (e.g., center of the desired time step interval).

Besides these state components, one might also want to consider some other factors, such as:

- **lanelet ID:** we can retrieve the ID of the lanelet on which the state is located. By this, we can determine whether the examined state is located on the lanelet of the goal state, and reward such states.
- **obstacles on lanelet:** contrary to the previous metric, if there are obstacles located on the lanelet of the goal state, we might want to make a lane change, thus penalizing such states.
- **trajectory efficiency:** we can calculate the ratio of the length of the trajectory traveled so far to the time required to travel such a trajectory as trajectory efficiency, and reward those trajectories with higher efficiencies.

Typically, these metrics are accompanied by different weights to collectively form the final heuristic function. The optimal weights are hard to determine; thus, we mostly have to make an initial guess and fine-tune them empirically. The ultimate heuristic function typically has the following form:

$$h(\text{state}) = \sum_{\text{i}} \text{weight}_{\text{i}} \times \text{metric}_{\text{i}}(\text{state})$$

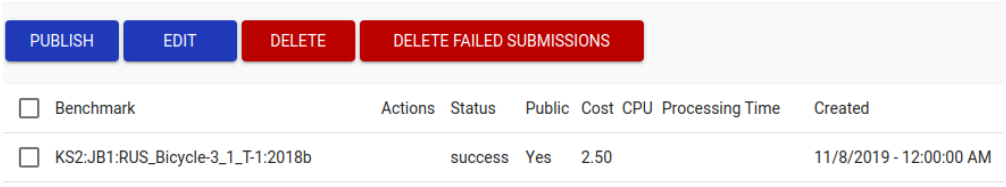
As a reference, you may find more explanation on planning algorithms in the book Planning Algorithms by Steven M. LaValle.

Additionally, a route planner package has been provided (see tutorial 5), which is able to generate feasible routes (and associated reference paths) that lead the initial lanelet(s) to the goal lanelet(s). One may incorporate additional information extracted from these routes and reference paths into the search algorithms to guide the search tree towards the goal.

7 Submitting Solutions

Your final solutions should be uploaded to the challenge webpage. Here are the steps of doing so:

1. Register here with your email address. Please provide real **First name** and **Last name** and choose **Username** freely.
2. Enroll in *AI Programming Exercise 2025 - Bonus exercise* here.
3. Click on **new submissions** and upload your solution .xml files. Wait for all submissions to be evaluated and verify their evaluation status on Username/My Submissions.
4. Select all submissions and **publish them**. Only submissions with status Success and Public will be counted!



5. Check your rank in the **leaderboard** on the challenge webpage.

Note: You still have to submit your planner and other required files to Moodle as explained in Sec. 8

7.1 Evaluation

The full benchmark ID is composed of three parts. Let us denote the IDs of the vehicle model, the cost function, the scenario, the version of the scenario, and the benchmark as letters M, C, S, V, and B, respectively. The benchmark ID is constructed by concatenating partial IDs with colons, which raises the following form:

$$\text{B} = \text{M:C:S:V}$$

After submitting the solutions to the CommonRoad website, they are evaluated via cost functions specified in the benchmark ID. The lower the cost, the higher the rank. In this exercise, we use M = `KS2` and C = `SM1`. You can check this paper and this repository for more detail about the vehicle model and the cost function.

8 Deadline and Passing Requirements

If you are a student of Fundamentals of Artificial Intelligence (IN2406) or Techniques in Artificial Intelligence (IN2062), you should submit your work on Moodle before **23.01.2026 12:00pm** CET. You need to submit following files:

1. Your own implementation of `search_algorithms/student.py`. It should be callable with the StudentMotionPlanner method in `motion_planner.py` and should generate your solution files. If you make use of other packages, make sure to also provide a `requirements.txt` file. We should be able to run your code on the virtual machine image (see Sec. 2.2) by installing the requirements and copy-pasting your `student.py` and other files (if any).
2. A compressed file containing all of the valid solutions which you have submitted to the CommonRoad website. (in **.zip format**)
3. Your configuration file for batch processing `batch_processing_config.yaml`.
4. Your motion primitive XML files, if you have created your own set of motion primitives.
5. When generating solutions, use **Kinematic Single-Track model (KS)**, **vehicle type BMW i320** (vehicle ID = 2), and **cost function ID SM1** (these are all set by default in the batch processing config file).

These items should be put directly under the submission folder on Moodle, i.e., not be put under sub-folders or be zipped further. Please **do not** use absolute paths in your code, but instead use relative paths to the working directory.

If you have questions regarding this exercise, please raise them in the corresponding Moodle Forum, as other students may share the same question.

9 Beyond the Limits: Prize Challenge

If you are ambitious enough, we recommend you to evaluate your planner on additional scenarios (you can download them from this link) and submit your solution files to our prize challenge (*AI Programming Exercise 2025 - Prize Challenge* here). A scenario selection page is available on the CommonRoad website. We have the following rewards for students with the most Top 1 solutions:

1. **Top 10**: invitation to a Pizza Night gathering with our group
2. **Top 3**: special certificate stating your outstanding performance
3. **Top 1**: €200 cash

Lastly, good luck with the exercise, and hope you enjoy :)